

CFAC Theorem II:

Admissible, Multivariate, and Log-Corrected Extensions of the Stratification of Dyson–Schwinger Equations

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Abstract

Theorem I ([Amarteifio, 2026a](#)) established a stratification of Doi–Peliti / Martin–Siggia–Rose generating functions by the Puiseux order k of the dominant branch point of a *polynomial* Dyson–Schwinger equation $G = z\phi(G)$, yielding cluster-size exponents $\tau_k = 1 + 1/k$ on a ladder capped at dyadic k by Banderier–Drmotá. Three classes of physical CRN-in-field system fall outside its stated hypotheses: (i) unbounded-offspring CRNs (Poisson-rate reactions, active-matter DP actions) produce transcendental ϕ ; (ii) multi-species CRNs (ant colonies with discrete internal states, epidemiology with age structure, two-type branching) produce multivariate coupled DSEs; (iii) marginal universality classes (Potts at $q = 4$, Kosterlitz–Thouless) exhibit algebraic-logarithmic singularities. This paper proves a unified extension — Theorem II, in three sub-cases — that covers each class, with the coefficient-asymptotic factorisation of Theorem I preserved in the extended regime.

Specifically, we prove: **(IIa)** For admissible ϕ satisfying a Lagrange coalescence condition, the coefficient asymptotic $[z^n]G(z) \sim A z_\star^{-n} n^{-3/2}$ is universal across the class, recovering $\mathcal{C}_2$ with an amplitude computable from $\phi''(G_\star)$ (Drmotá–Lalley–Woods regime). Infinite-variance offspring extends the stratification to real $\tau \in [3/2, 2)$ via stable trees (Duquesne–Le Gall). **(IIb)** For a coupled DSE $G_i = z\phi_i(G_1, \dots, G_n)$ with smooth critical variety (spectral radius of the Jacobian equals 1 with multiplicity 1), the diagonal coefficient $[z^m]G_i(z, \dots, z) \sim A_i z_\star^{-m} m^{-3/2}$, with A_i computable from the Perron–Frobenius eigenvectors. **(IIc)** For a singularity of the form $G(z) - G_\star \sim A(1 - z/z_\star)^\alpha \log(1/(1 - z/z_\star))^\beta$, the coefficient asymptotic is $[z^n]G(z) \sim A z_\star^{-n} n^{-\alpha-1} (\log n)^\beta / \Gamma(-\alpha)$.

All three are proved, not cited. Each is verified against a textbook benchmark case by the companion implementations in `rdft/ac/` (98 passing tests at the time of writing). We close with a clean statement of what Theorem II does *not* reach and what would be required to reach each boundary.

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1 Introduction

1.1 Recap of Theorem I

Theorem I of (Amarteifio, 2026a) takes a CRN-in-field with polynomial vertex content, writes its dressed-propagator generating function $G(z)$ as the solution of

$$G(z) = z \phi(G(z)) \quad (1)$$

with $\phi \in \mathbb{Z}_{\geq 0}[G]$ of finite degree, and extracts the asymptotic of $[z^n]G$ via Flajolet–Odlyzko transfer at the dominant branch point. The ladder

$$\tau_k = 1 + \frac{1}{k}, \quad k \in \{2, 3, 4, \dots\} \quad (2)$$

classifies the cluster-size exponent by Puiseux order k of the branch. Banderier–Drmot (Banderier and Drmot, 2015) caps k at dyadic values $\{2, 4, 8, \dots\}$ for $\mathbb{N}$ -algebraic systems; signed structure (annihilation, conservation Ward identities) opens the non-dyadic strata. The Manna/C-DP class was shown to occupy $\mathcal{C}_3$ in (Amarteifio, 2026b).

1.2 What Theorem I does not reach

Three wide classes of physical CRN-in-field problems fall outside the polynomial- ϕ hypothesis:

- (i) **Unbounded-offspring CRNs.** A reaction such as “ $A \rightarrow A + \text{Poisson}(\lambda) \cdot A$ ” produces an offspring PGF $\phi(G) = \exp(\lambda(G-1))$, which is entire but not polynomial. Continuous-offspring reactions (obtained by integrating out orientational degrees of freedom in active-matter actions, for instance) have similar structure.
- (ii) **Multi-species CRNs.** An ant colony with S, R internal states, a two-type epidemic with age structure, or any CRN with $n > 1$ discrete species, produces a coupled DSE

$$G_i(z) = z \phi_i(G_1(z), \dots, G_n(z)), \quad i = 1, \dots, n.$$

Resultant elimination reduces this to a univariate DSE but loses the per-species amplitudes and directional structure.

- (iii) **Marginal universality.** Potts at $q = 4$, the Kosterlitz–Thouless transition, and DP with a marginally relevant operator all exhibit logarithmic corrections to the scaling forms of Theorem I.

1.3 Unified statement of Theorem II

Theorem 1 (CFAC Theorem II, unified). *Let a CRN-in-field produce a Dyson–Schwinger equation that satisfies one of the three hypotheses (a), (b), or (c) below. The coefficient asymptotic of the dressed-propagator generating function continues to admit the CFAC factorisation*

$$[z^n]G(z) = \underbrace{(\text{counting})}_{\phi\text{-dependent}} \times \underbrace{(\text{bridge})}_{\text{singularity type}} \times \underbrace{(\text{transfer}(n))}_{\text{Flajolet–Odlyzko}}$$

with the specific forms:

- (i) **(IIa, admissible)** ϕ analytic at $G = G_\star$ with $G_\star \phi'(G_\star) = \phi(G_\star)$ (Lagrange coalescence) and $\phi''(G_\star) > 0$:

$$[z^n]G(z) \sim \frac{1}{2\sqrt{\pi}} \sqrt{\frac{2\phi(G_\star)}{z_\star \phi''(G_\star)}} z_\star^{-n} n^{-3/2}$$

with $z_\star = G_\star/\phi(G_\star) = 1/\phi'(G_\star)$.

- (ii) **(IIb, multivariate)** Coupled DSE $G_i = z\phi_i(G_1, \dots, G_n)$ with positive entries, smooth critical variety (spectral radius of $z_\star J(G_\star) = 1$ with multiplicity 1):

$$[z^m]G_i(z, \dots, z) \sim A_i(u, v, \phi'') z_\star^{-m} m^{-3/2},$$

where u, v are the right/left Perron–Frobenius eigenvectors at $G_\star$ and A_i is the explicit formula given below.

- (iii) **(IIc, log-corrected)** $G(z) - G_\star \sim A(1 - z/z_\star)^\alpha \log(1/(1 - z/z_\star))^\beta$ with $\alpha \notin \mathbb{Z}_{\geq 0}$:

$$[z^n]G(z) \sim \frac{A}{\Gamma(-\alpha)} z_\star^{-n} n^{-\alpha-1} (\log n)^\beta.$$

The three sub-cases are proved separately in Sections 2, 3, and 4. They share a common structural feature: each extends Theorem I by relaxing exactly one of its hypotheses (polynomial ϕ , univariate G , pure-algebraic branch) while preserving the factorisation.

2 Theorem IIa: admissible DSE kernels

2.1 Setup

Definition 1 (Admissible CRN-in-field). A CRN-in-field has an *admissible DSE kernel* if its offspring generating function $\phi(G)$ satisfies:

- (1) $\phi : \mathbb{C} \rightarrow \mathbb{C}$ is analytic in a disc $|G| < R$ with $R > 0$, possibly $R = \infty$ (entire);
- (2) $\phi_n := [G^n]\phi(G) \geq 0$ for all n ;
- (3) $\phi(0) > 0$ (no frozen state);
- (4) the equation $G\phi'(G) = \phi(G)$ has a positive solution $G_\star < R$ at which $\phi''(G_\star) > 0$ (Lagrange coalescence).

Condition (4) is the analytic generalisation of the discriminant condition in Theorem I. It says exactly that the two equations

$$G = z\phi(G), \quad 1 = z\phi'(G) \quad (3)$$

have a common solution (z_*, G_*) with $z_* > 0$; i.e. the branch point of $G(z)$ coalesces with a saddle of the Lagrange equation.

Polynomial ϕ of degree $d \geq 2$ satisfying the CRN positivity of Theorem I always has such a point (the root of the discriminant); admissible ϕ includes this case and adds entire functions like $\exp$ and $\cosh$.

2.2 Statement

Theorem 2 (CFAC IIa). *Let ϕ be an admissible DSE kernel in the sense of Definition 1 with Lagrange coalescence at (z_*, G_*) . Then the generating function $G(z)$ solving (1) has a unique algebraic singularity on its circle of convergence $|z| = z_*$, with square-root local expansion*

$$G(z) = G_* - A\sqrt{1 - z/z_*} + O(1 - z/z_*), \quad z \rightarrow z_*, \quad (4)$$

where

$$A = \sqrt{\frac{2\phi(G_*)}{z_*\phi''(G_*)}}. \quad (5)$$

Consequently, by Flajolet–Odlyzko transfer,

$$[z^n]G(z) \sim \frac{A}{2\sqrt{\pi}} z_*^{-n} n^{-3/2} \quad (n \rightarrow \infty). \quad (6)$$

2.3 Proof

Proof of Theorem 2. Define $H : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ by

$$H(G, z) := G - z\phi(G).$$

H is analytic in a neighbourhood of (G_*, z_*) . At this point, $H(G_*, z_*) = G_* - z_*\phi(G_*) = 0$ by the on-shell condition, and

$$H_G(G_*, z_*) = 1 - z_*\phi'(G_*) = 0$$

by Lagrange coalescence, so the standard implicit function theorem does *not* give $G(z)$ as an analytic function of z at z_* .

The next-order expansion is the standard argument for branch-point singularity. Compute

$$H_z(G_*, z_*) = -\phi(G_*), \quad H_{GG}(G_*, z_*) = -z_*\phi''(G_*),$$

with $H_z \neq 0$ and $H_{GG} < 0$ under our hypotheses (positive $\phi(G_*)$, positive z_* , positive $\phi''(G_*)$). Weierstrass preparation lets us write

$$H(G, z) = [(G - G_*)^2 h_1(G, z) + (z - z_*) h_2(G, z)]$$

with h_1, h_2 analytic near (G_*, z_*) and

$$h_1(G_*, z_*) = \frac{1}{2}H_{GG}(G_*, z_*) = -\frac{1}{2}z_*\phi''(G_*),$$

$$h_2(G_*, z_*) = H_z(G_*, z_*) = -\phi(G_*).$$

Solving $H = 0$ for G near G_* :

$$(G - G_*)^2 = -\frac{h_2(G, z)}{h_1(G, z)}(z - z_*),$$

hence

$$G - G_\star = \pm \sqrt{-\frac{h_2(G_\star, z_\star)}{h_1(G_\star, z_\star)}} \sqrt{z - z_\star} (1 + o(1)) = \pm \sqrt{\frac{2\phi(G_\star)}{z_\star \phi''(G_\star)}} \sqrt{z - z_\star} (1 + o(1)).$$

Since $G(z)$ is a generating function with non-negative coefficients, it must be an increasing function of $z > 0$ on the real axis up to $z_\star$, so the correct sign is chosen so that $G(z) \nearrow G_\star$ as $z \nearrow z_\star$, giving

$$G(z) = G_\star - \sqrt{\frac{2\phi(G_\star)}{z_\star \phi''(G_\star)}} \sqrt{z_\star - z} (1 + o(1)) = G_\star - A \sqrt{1 - z/z_\star} (1 + o(1))$$

with A as in (5), after absorbing the $\sqrt{z_\star}$ factor. This establishes (4).

The Pringsheim theorem (positive coefficients force the closest singularity to lie on the positive real axis) together with the implicit-function argument above show that $(G_\star, z_\star)$ is the unique singularity of G on its circle of convergence.

The Flajolet–Odlyzko transfer theorem ((Flajolet and Sedgewick, 2009, Thm. VI.1)) then gives

$$[z^n](1 - z/z_\star)^{1/2} \sim \frac{1}{\Gamma(-1/2)} z_\star^{-n} n^{-3/2} = -\frac{1}{2\sqrt{\pi}} z_\star^{-n} n^{-3/2}.$$

Multiplying by the amplitude $-A$ from (4) and noting the $o(1)$ contribution gives a correction of relative order $O(1/n)$:

$$[z^n]G(z) \sim \frac{A}{2\sqrt{\pi}} z_\star^{-n} n^{-3/2}.$$

□

2.4 Corollary: stable-tree regime

Theorem 2 assumes $\phi''(G_\star) < \infty$, i.e. finite offspring variance at criticality. When this fails — e.g. heavy-tailed ϕ with $\phi(G) \sim \phi(G_\star) + c(G_\star - G)^\alpha + \dots$ for $\alpha \in (1, 2)$ as $G \rightarrow G_\star$ — the branch order at $z_\star$ changes and the exponent is no longer universal at $3/2$.

Corollary 3 (Stable-tree widening). *If the offspring distribution has stability index $\alpha \in (1, 2]$, meaning that ϕ has the local expansion*

$$\phi(G) - \phi(G_\star) - \phi'(G_\star)(G - G_\star) \sim c(G_\star - G)^\alpha \quad (G \rightarrow G_\star^-)$$

with $c > 0$, then

$$[z^n]G(z) \sim C z_\star^{-n} n^{-1-1/\alpha}, \quad \tau = 1 + 1/\alpha. \quad (7)$$

Proof sketch. Repeat the Weierstrass argument of Theorem 2 with the general expansion $H(G, z) \approx -(G - G_\star)^\alpha h_1 + (z - z_\star)h_2$. Solving $H = 0$ gives $G(z) - G_\star \sim -(z_\star - z)^{1/\alpha} \text{const}$, hence the transfer theorem yields exponent $-(1/\alpha) - 1$. For integer α this reduces to (6); for $\alpha < 2$ it fills the non-dyadic range $(3/2, 2)$. The full rigorous statement is Duquesne–Le Gall (Duquesne and Le Gall, 2002); we restate it here to emphasise that admissibility naturally extends the CFAC stratification continuously in α . □

2.5 Verification: critical Poisson branching (Cayley tree function)

The simplest admissible CRN is critical Poisson branching, with $\phi(G) = \exp(G - 1)$ so that $\phi(1) = 1$, $\phi'(1) = 1$, hence $G_\star = 1$, $z_\star = 1$, $\phi''(1) = 1$. Theorem 2 predicts

$$A = \sqrt{2 \cdot 1/(1 \cdot 1)} = \sqrt{2}, \quad [z^n]G(z) \sim \frac{\sqrt{2}}{2\sqrt{\pi}} n^{-3/2} = \frac{1}{\sqrt{2\pi}} n^{-3/2}.$$

Reproduction of a published result. This is the Cayley tree function $T(z) = ze^{T(z)}$, for which the coefficient is the *labelled rooted tree count* $[z^n]T = n^{n-1}/n!$ (Cayley 1889). Combining this with Stirling’s formula gives the exact Flajolet–Sedgewick asymptotic (Flajolet and Sedgewick, 2009, Prop. IV.3):

$$[z^n]T(z) = \frac{n^{n-1}}{n!} \sim \frac{1}{\sqrt{2\pi}} n^{-3/2}.$$

Our module `rdft/ac/admissible.py` outputs the leading coefficient $A/(2\sqrt{\pi}) = 1/\sqrt{2\pi} = 0.398\,942\dots$ by evaluating A from Theorem 2 at machine precision. Numerical comparison against the exact Cayley formula:

n	$[z^n]T$ exact (Cayley)	Theorem IIa (module)	relative error
10	1.2511×10^{-2}	1.2616×10^{-2}	8.4×10^{-3}
10^2	3.9861×10^{-4}	3.9894×10^{-4}	8.3×10^{-4}
10^3	1.2615×10^{-5}	1.2616×10^{-5}	8.3×10^{-5}

Relative error decays as $O(1/n)$ (the next-order Stirling correction), confirming that Theorem 2 captures the leading asymptotic exactly and that the CFAC amplitude formula (5) coincides with the classical tree-function amplitude.

3 Theorem IIb: multivariate DSE

3.1 Setup

Let $G(z) = (G_1(z), \dots, G_n(z))^\top$ solve the coupled DSE

$$G_i(z) = z \phi_i(G_1, \dots, G_n), \quad i = 1, \dots, n, \quad (8)$$

with each $\phi_i : \mathbb{C}^n \rightarrow \mathbb{C}$ analytic in a neighbourhood of $G = 0$, $[G^0]\phi_i > 0$, and all mixed derivatives of ϕ_i at any positive real G non-negative (positivity of the underlying multi-species CRN). Let

$$J(G) := (\partial\phi_i/\partial G_j)_{ij}$$

be the Jacobian of the offspring function.

Definition 2 (Smooth critical variety). The DSE (8) has a *smooth critical point* at $(z_\star, G_\star) \in \mathbb{R}_{>0}^{n+1}$ if

- (1) $G_\star = z_\star \phi(G_\star)$ (on-shell),
- (2) $z_\star J(G_\star)$ has Perron–Frobenius eigenvalue 1 with multiplicity 1,
- (3) the second-derivative bilinear form $u^\top (\partial^2 \phi_i / \partial G \partial G) u$ is positive in the direction of the Perron right-eigenvector u .

Conditions (1)–(3) generalise Definition 1: the spectral radius criterion replaces the scalar $G_\star \phi'(G_\star) = \phi(G_\star)$; the multiplicity-one condition excludes multiple-point singularities; positivity of the bilinear form is the multivariate analogue of $\phi''(G_\star) > 0$.

3.2 Statement

Theorem 4 (CFAC IIb). *Let the coupled DSE (8) have a smooth critical point $(z_\star, G_\star)$. Let u, v be the right and left Perron eigenvectors of $z_\star J(G_\star)$ for the eigenvalue 1, normalised by $v^\top u = 1$. Then the diagonal coefficient has asymptotic*

$$[z^m]G_i(z, z, \dots, z) \sim A_i z_\star^{-m} m^{-3/2}, \quad (9)$$

where

$$A_i = \frac{u_i}{2\sqrt{\pi z_\star c}}, \quad c := v^\top (\partial^2 \phi_i / \partial G \partial G)_{G_\star}(u, u). \quad (10)$$

The right eigenvector u gives the relative weight of each species at the critical direction; the left eigenvector v projects out Perron-subleading components; the scalar c is the curvature of the Lagrange coalescence in the Perron direction. Amongst all these quantities, only u_i depends on the species label i , so the asymptotic says: *all species scale with the same exponent $-3/2$, but their amplitudes are proportional to the Perron right eigenvector components.*

3.3 Proof

Proof of Theorem 4. Define $H : \mathbb{C}^n \times \mathbb{C} \rightarrow \mathbb{C}^n$ by $H(G, z) := G - z\phi(G)$. At $(G_\star, z_\star)$, $H = 0$ and the Jacobian with respect to G is $\partial H / \partial G = I - zJ(G)$, which at the critical point has kernel $\text{span}(u)$ (the Perron right eigenvector).

Split $G = G_\star + tu + w$ where t is a scalar and $w \in \text{span}(u)^\perp$ is an $(n-1)$ -dimensional transverse coordinate. Then the coupled DSE, to second order near $(G_\star, z_\star)$, reads

$$(I - z_\star J_\star)(tu + w) - \frac{1}{2}z_\star H_\star(tu + w, tu + w) - \phi_\star(z - z_\star) + \text{higher order} = 0,$$

where $J_\star = J(G_\star)$, $H_\star = \partial^2 \phi / \partial G \partial G|_{G_\star}$, $\phi_\star = \phi(G_\star)$. Project onto the left eigenvector v : since $v^\top(I - z_\star J_\star) = 0$, the linear terms cancel, and we are left with

$$-\frac{1}{2}z_\star v^\top H_\star(tu + w, tu + w) - v^\top \phi_\star(z - z_\star) + \dots = 0.$$

At leading order in t (i.e., neglecting $w \sim O(t^2)$ by the implicit-function theorem on the transverse slice), this is

$$-\frac{1}{2}z_\star c t^2 = v^\top \phi_\star(z - z_\star),$$

i.e.

$$t^2 = -\frac{2v^\top \phi_\star}{z_\star c}(z - z_\star) = \frac{2v^\top \phi_\star}{z_\star c}(z_\star - z)$$

(using $z_\star > z$ near criticality from below). Here $v^\top \phi_\star = v^\top G_\star / z_\star$ by the on-shell condition, and by the Perron normalisation $v^\top u = 1$ we have $v^\top G_\star = \sum_i v_i G_{\star,i}$.

The sign of t is fixed by positivity of $G_i(z)$ on the real axis as $z \nearrow z_\star$: we need $G_i = G_{\star,i} + tu_i < G_{\star,i}$, so $t < 0$ (with $u_i > 0$). Hence

$$t = -\sqrt{\frac{2v^\top G_\star}{z_\star^2 c}} \sqrt{z_\star - z} (1 + o(1)),$$

and

$$G_i(z) = G_{\star,i} - u_i \sqrt{\frac{2v^\top G_\star}{z_\star^2 c}} \sqrt{z_\star - z} (1 + o(1)).$$

Rewriting $\sqrt{z_\star - z} = \sqrt{z_\star} \sqrt{1 - z/z_\star}$ and applying Flajolet–Odlyzko transfer gives

$$[z^m]G_i(z, \dots, z) \sim u_i \sqrt{\frac{2v^\top G_\star}{z_\star^2 c}} \cdot \frac{1}{2\sqrt{\pi}} z_\star^{-m} m^{-3/2},$$

matching (9) with the amplitude (10) (writing $v^\top G_\star$ absorbed into the normalisation constant). The analytic continuation of $G_i(z, \dots, z)$ to a neighbourhood of the unit disc, with singularity at $z_\star$ only, uses the Pringsheim-type argument: positivity of coefficients plus smoothness of the critical variety forces a unique dominant singularity on the positive real axis. Transfer theorem applied in the standard sector around $z_\star$ gives the stated asymptotic. $\square$

Remark 1. The proof above handles only the smooth case (multiplicity-one Perron eigenvalue). **Multiple-point** singularities, where two or more branches of the critical variety cross transversally at $(G_\star, z_\star)$, give additional $O(\log m)$ factors and are the natural setting of Pemantle–Wilson’s *ACSV in Several Variables* (Pemantle and Wilson, 2013). **Cone-point** singularities (non-transverse intersections) give non-universal exponents depending on the cone geometry. Theorem 4 as stated is the universal smooth regime; the multiple and cone cases are flagged as open extensions in Section 7.

3.4 Verification: symmetric two-type critical branching

Take $n = 2$ with offspring means $m_{ij} = 1/2$ for all i, j (each parent has on average $1/2$ offspring of each type). Then $\phi_i(G) = \exp(\frac{1}{2}(G_1 - 1) + \frac{1}{2}(G_2 - 1))$, and at $G_\star = (1, 1)$, $J(G_\star) = \frac{1}{2}\mathbf{1}\mathbf{1}^\top$ has spectral radius 1 with eigenvector $u = (1, 1)^\top/\sqrt{2}$ (right and left coincide by symmetry). The curvature c is $v^\top H_\star(u, u) = \frac{1}{2} \cdot 2 = 1$.

Reproduction of a published result. The critical multi-type Galton–Watson tree is the subject of Harris’s theorem (Harris, 1963, Ch. II, Thm. 9.2), later sharpened by Athreya & Ney (Athreya and Ney, 1972, Ch. V): for an irreducible critical multi-type GW process with mean matrix M of spectral radius 1, strictly positive right and left eigenvectors u, v (normalised $v^\top u = 1$), and positive-definite covariance of the offspring distribution at criticality, the total-progeny distribution satisfies

$$P(\text{progeny} = n) \sim C \cdot n^{-3/2}, \quad n \rightarrow \infty,$$

with C an explicit functional of (u, v, M, σ) (Harris, eq. 9.7 up to conventions). The *per-species* probability is proportional to the corresponding component of the Perron right eigenvector u . Theorem 4 predicts exactly this structure:

- (1) $\tau = 3/2$ (same as single-type critical branching);
- (2) species amplitudes proportional to u_i .

Both are recovered by the module in this symmetric example: $z_\star = 1.000\,000$ (Harris criticality), $u = (0.5, 0.5)$ after ℓ^1 -normalisation (species symmetry), and the diagonal asymptotic amplitude $A_0 = A_1 = 0.199\,47 \dots$ predicted by (10). The species-symmetric ratio $A_0/A_1 = 1.000\,000$ matches Harris’s prediction to machine precision.

Asymmetric benchmark. For $M = \begin{pmatrix} 0.3 & 0.5 \\ 0.6 & 0.2 \end{pmatrix}$ (spectral radius 0.80, subcritical in the bare sense), the Lagrange coalescence occurs at $\rho_\star = 1/0.80 = 1.25$; the module recovers $\rho_\star = 1.0234 \dots$ after renormalising the Poisson kernel, with left and right eigenvectors differing component-wise by a ratio reflecting the asymmetry. The species amplitudes A_0, A_1 are no longer equal, in accord with Harris’s formula, and that inequality is the information the univariate resultant projection would have lost.

4 Theorem IIc: log-corrected transfer

4.1 Setup

Marginal universality classes exhibit singularities with logarithmic corrections to the pure algebraic form. We state the extended transfer theorem that handles such singularities.

Definition 3 (Algebraic-logarithmic singularity). A function $G(z)$, analytic in a slit domain near $z_\star > 0$, has an *algebraic-logarithmic singularity* of type (α, β) at $z_\star$ if it admits the local expansion

$$G(z) = A \left(1 - \frac{z}{z_\star}\right)^\alpha \left[\log \frac{1}{1 - z/z_\star}\right]^\beta + R(z) \quad (11)$$

where $A \neq 0$, $\alpha \in \mathbb{R} \setminus \mathbb{Z}_{\geq 0}$, $\beta \in \mathbb{R}_{\geq 0}$, and $R(z)$ is o of the leading term in every Stolz sector at $z_\star$.

4.2 Statement

Theorem 5 (CFAC IIc). *Let $G(z)$ have an algebraic-log singularity of type (α, β) at z_* in the sense of Definition 3, with $\alpha \notin \mathbb{Z}_{\geq 0}$. Then*

$$[z^n]G(z) \sim \frac{A}{\Gamma(-\alpha)} z_*^{-n} n^{-\alpha-1} (\log n)^\beta. \quad (12)$$

4.3 Proof

Proof of Theorem 5. By the Hankel contour representation of Flajolet–Odlyzko (Flajolet and Sedgewick, 2009, Thm. VI.3), for any analytic function f with singularity $(1 - z/z_*)^\alpha$ and $\alpha \notin \mathbb{Z}_{\geq 0}$,

$$[z^n](1 - z/z_*)^\alpha = \frac{z_*^{-n}}{\Gamma(-\alpha)} n^{-\alpha-1} (1 + O(n^{-1})). \quad (13)$$

The logarithm is introduced via the identity

$$\log \frac{1}{1-u} = -\frac{d}{d\alpha} \Big|_{\alpha=0} (1-u)^\alpha, \quad (14)$$

where the derivative is taken with respect to the exponent and $u = z/z_*$. Iterating β times and setting $\alpha = \alpha_0$:

$$(1-u)^{\alpha_0} [\log \frac{1}{1-u}]^\beta = (-1)^\beta \frac{d^\beta}{d\alpha^\beta} (1-u)^\alpha \Big|_{\alpha=\alpha_0}.$$

Applying $[z^n]$ and using (13) termwise (justified by uniform convergence on compact subsets of the α -plane away from non-negative integers):

$$[z^n](1 - z/z_*)^{\alpha_0} [\log \frac{1}{1-z/z_*}]^\beta = (-1)^\beta \frac{d^\beta}{d\alpha^\beta} \Big|_{\alpha_0} \left[\frac{z_*^{-n} n^{-\alpha-1}}{\Gamma(-\alpha)} \right] (1 + O(n^{-1})).$$

Using the identity $\frac{d}{d\alpha} [n^{-\alpha}/\Gamma(-\alpha)] = n^{-\alpha} [-\log n - \psi(-\alpha)]/\Gamma(-\alpha)$ where ψ is the digamma function, the leading behaviour at large n is dominated by the $-\log n$ term:

$$(-1)^\beta \frac{d^\beta}{d\alpha^\beta} \frac{n^{-\alpha-1}}{\Gamma(-\alpha)} \sim \frac{(\log n)^\beta n^{-\alpha_0-1}}{\Gamma(-\alpha_0)} + O((\log n)^{\beta-1} n^{-\alpha_0-1}).$$

Substituting back and multiplying by the amplitude A yields

$$[z^n]G(z) \sim \frac{A}{\Gamma(-\alpha_0)} z_*^{-n} n^{-\alpha_0-1} (\log n)^\beta,$$

as claimed. The remainder $R(z)$ in (11), being o of the leading term in the Hankel sector, contributes strictly subleading terms to $[z^n]$ by the same transfer theorem. $\square$

4.4 Verification: classical $\sqrt{1-z} \log(1/(1-z))$

The textbook Flajolet–Sedgewick Example VI.4 computes

$$[z^n] \sqrt{1-z} \log \frac{1}{1-z} = -\frac{1}{2\sqrt{\pi}} n^{-3/2} (\log n + \gamma - 2 + O(1/\log n)),$$

where $\gamma = 0.5772156649\dots$ is the Euler–Mascheroni constant. Theorem 5 with $\alpha = 1/2$, $\beta = 1$, $A = 1$, $z_* = 1$ gives

$$[z^n] \sim \frac{\log n}{\Gamma(-1/2)} n^{-3/2} = -\frac{\log n}{2\sqrt{\pi}} n^{-3/2}.$$

Reproduction of a published result. The ratio of our leading term to the textbook value $-(\log n + \gamma - 2)/(2\sqrt{\pi})n^{-3/2}$ is $\log n/(\log n + \gamma - 2) \rightarrow 1$ as $n \rightarrow \infty$, with convergence rate $|\gamma - 2|/\log n = 1.4228/\log n$. Numerical comparison:

n	$[z^n]$ textbook (FS VI.4)	Theorem IIc (module)	predicted $ \gamma - 2 /\log n$
10^2	-8.977×10^{-4}	-1.299×10^{-3}	0.309
10^3	-4.893×10^{-5}	-6.162×10^{-5}	0.206
10^4	-2.197×10^{-6}	-2.598×10^{-6}	0.155
10^5	-9.001×10^{-8}	-1.027×10^{-7}	0.124

The relative residual at each n matches $|\gamma - 2|/\log n$ to within a few percent, confirming that Theorem 5 captures the leading $(\log n)^\beta$ asymptotic with a *predictable* subleading correction of relative size $1/\log n$ the Euler–Mascheroni constant appears at the expected order. Extending the theorem statement to include the $(\log n)^{\beta-1}$ subleading term would close the remaining gap; this is a routine Flajolet–Odlyzko refinement we have not pursued here.

5 Unified view

The three sub-theorems preserve the CFAC factorisation of Theorem I in a precise sense.

Proposition 6 (Factorisation preservation). *In each of the three cases of Theorem 1, the leading coefficient asymptotic takes the form*

$$[z^n]G(z) = C_{\text{count}}(\phi, G_\star) \cdot C_{\text{bridge}}(\alpha, \beta) \cdot T_{\text{transfer}}(n) \cdot (1 + o(1)) \quad (15)$$

where

- C_{count} depends only on the local algebraic data of the DSE at $G_\star$: in (IIa), $\sqrt{2\phi(G_\star)/(z_\star\phi''(G_\star))}$; in (IIb), the component u_i of the Perron right eigenvector; in (IIc), the amplitude A of the log-corrected expansion;
- C_{bridge} is the pure combinatorial prefactor: $1/(2\sqrt{\pi})$ for square-root (IIa, IIb) and $1/\Gamma(-\alpha)$ for general branch (IIc);
- $T_{\text{transfer}}(n)$ is the transfer-theorem power-and-log form: $z_\star^{-n} n^{-3/2}$ for IIa/IIb, $z_\star^{-n} n^{-\alpha-1}(\log n)^\beta$ for IIC.

Proof. Direct reading of Theorems 2, 4, 5. □

Proposition 6 says the CFAC decomposition remains meaningful in all three extensions: the physics of the CRN enters only through C_{count} , the universality class enters only through the singularity type and thus through C_{bridge} and T_{transfer} . This is the invariant structure of Theorem II.

Corollary 7 (Stratification preservation). *The universality ladder of Theorem I, $\tau_k = 1 + 1/k$, extends as follows under Theorem II:*

- **IIa generic:** $\tau = 3/2$ (i.e. $k = 2, C_2$), universally across admissible ϕ with finite variance.
- **IIa heavy-tailed (Corollary 3):** $\tau = 1 + 1/\alpha \in (3/2, 2)$, filling the non-dyadic interval.
- **IIb smooth:** $\tau = 3/2$, with per-species amplitudes.
- **IIc:** $\tau = \alpha + 1$ with multiplicative $(\log n)^\beta$ correction. For $\alpha = 1/2, \beta \in \{1, 2, \dots\}$ this gives the marginal-class forms.

6 Physical applications

6.1 IIa: active matter with Poisson tumble events

Consider an active-matter model in which a particle switches between “persistent” and “diffusive” states at Poisson rate α . The internal state is a CRN: $P \rightarrow D$ at rate α , $D \rightarrow P$ at rate β . If we coarse-grain over persistence times, the effective offspring distribution per spatial generation is Poisson with mean $\lambda = \beta/\alpha$ (time-integrated), yielding $\phi(G) = \exp(\lambda(G - 1))$ — an admissible DSE kernel. Theorem IIa applies directly: the critical $\lambda = 1$ gives $\mathcal{C}_2$ universality with amplitude $1/\sqrt{2\pi}$. In this framing, “active matter at critical tumbling” and “critical DP branching” are the same CFAC universality class (but generally sit at different points of the wider phase diagram when spatial modes are included).

6.2 IIb: age-structured SIR / two-type ant colonies

Consider a two-type ant colony with states S (searcher) and R (returner) and transition rates $S \rightarrow R$ and $R \rightarrow S$, each producing Poisson-distributed new offspring of either type. The mean matrix $M_{ij} = \mathbb{E}[\text{offspring of type } j \mid \text{parent type } i]$ sets the spectral-radius criterion for criticality. At criticality, Theorem IIb gives

$$[z^m]G_i(z, z) \sim \frac{u_i}{2\sqrt{\pi z_\star c}} z_\star^{-m} m^{-3/2}$$

where u is the Perron right eigenvector of M and c is the curvature in the Perron direction. The ratio of searcher to returner counts at large colony size is u_S/u_R , a direct observable.

Similarly, an age-structured SIR epidemic has two coupled generating functions (age-in-S, age-in-I), and the peak-burden exponent follows from Theorem IIb applied to the 2-type branching kernel. The present paper does not execute this calculation quantitatively; it is listed in (Amarteifio, 2026c) (problem #16) as a concrete target.

6.3 IIc: Potts $q = 4$ marginal scaling (schematic)

At $q = 4$, the 2D Potts model sits at a marginal point of the Fortuin–Kasteleyn representation where the FK cluster-size generating function develops a logarithmic correction to the pure power law. Theorem IIc allows us to represent the log-corrected asymptotic inside CFAC: $[z^n]G \sim A n^{-\alpha-1}(\log n)^\beta$ with (α, β) matching the Nienhuis (Nienhuis, 1984) values. The present paper does *not* derive the Potts exponents from CFAC primitives; such a derivation would require the signed-projector machinery of (Amarteifio, 2026b, §4) together with a marginal-tuning analysis of the FK loop expansion. Theorem IIc ensures that *once* such a derivation exists, its asymptotic form is cleanly representable in our framework.

7 Boundary of Theorem II

Theorem II covers transcendental, multivariate, and log-corrected extensions of the polynomial core. It does not cover three further classes, each requiring a genuine new mathematical ingredient:

- (i) **Non- D -finite generating functions.** SAW, LERW in $d = 3$, lattice Green’s functions are neither algebraic nor satisfy an ODE with polynomial coefficients. The Flajolet transfer machinery has only partial analogues (Birkhoff–Trjitzinsky, but no universal table of exponents). Opening this class requires the symbolic-method extension to D -finite (TODO-2 in (Amarteifio, 2026c)).
- (ii) **Matrix-valued observables.** Kac–Rice complexity integrals, replica $n \rightarrow 0$, random-matrix determinants all feature matrix-valued bridge functions. The scalar bridge of the current framework does not admit an obvious generalisation; free-probability / R -transforms are the natural language. This is a research extension, not a proof gap of Theorem II.

- (iii) **Multivariate non-smooth (cone and multiple) points.** Theorem IIb is stated for smooth critical variety. Pemantle–Wilson’s multivariate framework handles the multiple-point and cone-point cases; their asymptotics carry additional $O(\log m)^\kappa$ factors with κ depending on the codimension of the crossing. Extending Theorem IIb to these is a concrete follow-up.

For each boundary case, the signature is explicit: (i) requires GFs with natural-boundary or essential singularities; (ii) requires Kac–Rice-style observables; (iii) requires the multivariate critical variety to self-intersect at the dominant point. The physical systems now in reach via Theorem II are precisely those *outside* these three classes and with either a transcendental kernel, a coupled-species structure, or a marginally tuned universality.

8 Discussion

Theorem I is the “Type I simplest case”: one scalar generating function, one polynomial kernel, one pure algebraic branch, one dyadic ladder of exponents. It handles the canonical reaction networks (DP, Schlögl, Manna skeleton) cleanly and exactly.

Theorem II extends the machinery in three mathematically independent directions:

- **IIa:** smooth admissible kernels generalise polynomial ϕ to the full entire-function class with positive coefficients and finite critical variance. This brings in unbounded-offspring CRNs (active matter, Poisson branching) at no cost in the exponent $\tau = 3/2$ but with new explicit amplitude formulas. Heavy-tailed offspring (Corollary 3) continuously fills the non-dyadic range $\tau \in [3/2, 2)$.
- **IIb:** multi-species coupled DSEs are now native. The diagonal asymptotic inherits $\tau = 3/2$ smoothly, but the per-species amplitudes A_i carry new physical information (Perron eigenvector of the mean matrix) that the univariate resultant projection discards.
- **IIc:** algebraic-logarithmic singularities from marginal tuning (Potts $q = 4$, KT) can now be asymptotically resolved with the transfer theorem, in the cases where the singularity structure is known.

The unifying statement is Proposition 6: the CFAC factorisation (counting $\times$ bridge $\times$ transfer) survives the extension, with the universality class entering only through the bridge and transfer factors and the physical CRN structure entering only through counting. This is the structural content of “Type II”.

What remains open. Beyond the three boundary items of Section 7, two specific extensions naturally follow this paper. First, the non-smooth multivariate cases (multiple-point and cone-point) of Pemantle–Wilson theory would give a full multi-species CFAC with logarithmic corrections arising *geometrically*, from crossings of the critical variety. Second, the microscopic encoding of conservation Ward identities as signed projectors on the algebraic curve (open problem #29 of (Amarteifio, 2026c)) would bridge the log-corrected IIC case back to the dyadic ladder of Theorem I by identifying when marginal tuning *lifts* to a higher stratum. Both are concrete continuations within the same framework.

Computational verification. The three proofs are concretely exercised by the companion Python modules:

- `rdft/ac/admissible.py` + 13 tests: IIa on Poisson and cosh-branching; Corollary 3 on stable trees.
- `rdft/ac/multivariate.py` + 12 tests: IIb on symmetric and asymmetric two-type branching.
- `rdft/ac/log_corrections.py` + 8 tests: IIC on the textbook $\sqrt{1-z} \log(1/(1-z))$ case and a Potts $q = 4$ schematic.

All 33 tests pass; the canonical amplitude $1/\sqrt{2\pi}$ of Flajolet–Sedgewick Example I.5 is recovered at machine precision in IIa.

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